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Cultural and Creative Cities Monitor

2023 Update

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Abstract

The Cultural and Creative Cities Monitor (hereafter ‘the monitor’) is a monitoring and benchmarking tool that was first developed in 2017 to allow European cities to compare and contrast their areas of excellence and improvement in terms of culture and creativity. Since it was first launched, the monitor has served as a guidance tool for policymakers at the local, national and European levels.

The 2023 update of the monitor involved a methodological improvement and a revision of selected indicators, to ensure that the tool remains reliable and coherent, enabling meaningful comparisons of cities’ performance over time. The monitor’s latest results are available for three different reference years, with a focus on the most recent year, 2019. These results provide a snapshot of the situation just before the COVID-19 pandemic.

The updated statistical assessment of the monitor described in this report gives the user an understanding of how to use the monitor, enabling the informed use of the data and monitoring tools provided.

The monitor’s online platform has also been updated. In addition to the pre-existing functionalities, the online tool now also allows users to compare and contrast the performance of 196 European cities at three different points in time.

Acknowledgements

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In addition, the authors would like to thank colleagues and former colleagues from the European Commission's Directorate-General for Education, Youth, Sport and Culture (Inna Garkova, Barbara Stacher, Marta Jimenez Pumares and Catherine Magnant), who took part in open and constructive exchanges throughout the years to both improve the monitor and steer its future direction.

Lastly, the authors would like to thank all national, regional and local authorities that have been using the monitor over the last few years and have provided valuable feedback and input, driving the evolution of the tool.

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1. Introduction

The first edition of the Cultural and Creative Cities Monitor (hereafter ‘the monitor’) was released in 2017, covering a sample of 168 European cities of different sizes. Subsequently, in 2019, the sample of cities was increased to 190. The development of the monitor was requested and supported by Commissioner Tibor Navracsics (Juncker Commission), to provide a benchmarking tool for culture and creativity in Europe at the city level. A thorough review of 40 indices related to cities’ culture and creativity helped to identify gaps that a new measurement and benchmarking tool could aim to fill. The result is a toolkit that provides comparable data at the city level for a wide sample of cities across Europe. The monitor aggregates metrics related to the cultural and creative sectors of each city and allows users to compare the relative strengths of similarly sized cities in these domains.

The monitor is made up of three components: this technical report, an online tool⁽¹⁾ and an annex containing the data used to construct the monitor itself⁽²⁾. The 2023 update of the monitor provides a synthetic measure of culture and creativity in 196 cities in Europe for three reference years: 2015, 2017 and 2019.

Since its first edition, the monitor has been showcased in more than 30 cities and has helped local administrations to identify areas of unexplored potential in their cities. It has been used as a benchmarking and peer-learning tool by a number of European cities, and has fed into EU policy documents, including the new European agenda for culture⁽³⁾.

Since the 2017 edition, the monitor has looked at how cities are able to provide local residents with cultural amenities, explored the ability of cities to attract visitors and attract and retain talent, and produced innovative outputs. Moreover, the monitor has provided an overview of the performance of European cities in areas that support the cultural and creative sectors, such as the openness of their populations and the quality of their transport connections and regional governance.

While the 2023 update of the monitor aims to continue the mapping and benchmarking exercises initiated in 2017, the situation has profoundly changed since then. At the time of writing, data to perform a meaningful analysis of the impacts of and ability to recover from the COVID-19 pandemic’s effects on culture and creativity in European cities were not available for a sufficient number of cities in the sample. Nevertheless, the present update of the monitor provides a pre-pandemic reference point (as it is based on data from 2019) that could be used to analyse the effects of the COVID-19 pandemic on European cultural and creative cities once data for the post-pandemic period are available.

The 2023 update of the monitor brings some important methodological improvements. First, it allows for comparisons of cities’ performance over time. In this update, benchmarking is possible not only vis-à-vis peer cities with similar populations, gross domestic product (GDP) per capita and employment rates, but also for the same city over time. The aim is to help cities to identify opportunities and challenges, and to monitor their progress over time. Second, some amendments have been made in terms of data gathering and imputation methodology, thus allowing for more meaningful comparisons over time. While both previous editions of the monitor (2017 and 2019) used the latest data available for each city and indicator, bringing together observed data for multiple years, the revised approach uses data from only the reference years (i.e. 2015, 2017 and 2019) and imputes missing data, thus allowing for more meaningful comparisons over time.

This technical report has three main chapters: Chapter 2 ‘Methodology’ provides an overview of the methodology used for the development of the monitor and the main changes implemented since the 2019 edition; Chapter 3 ‘Statistical assessment’ provides an assessment of the 2023 update; and Chapter 4 ‘Online platform’ provides a quick guide to the online tool. The methodology chapter is intended to aid the accurate interpretation of the monitor’s results, the statistical assessment chapter is designed to allow a more specialised audience to explore the details and limitations of the monitor, and Chapter 4 offers support to users of the monitor’s accompanying online tool.

(1) <https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor>.

(2) <https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor/docs-and-data>.

(3) European Parliament resolution of 11 December 2018 on the new European agenda for culture (2018/2091(INI)) (<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52018IP0499>).

2. Methodology

2.1. Conceptual framework

The monitor is a composite indicator and was developed based on the methodology outlined in the *Handbook on Constructing Composite Indicators – Methodology and user guide* (OECD and JRC, 2008). Since its first edition (2017), the monitor has revolved around three main concepts – or subindices – considered relevant when attempting to measure a city’s cultural and creative liveliness: Cultural Vibrancy, Creative Economy and Enabling Environment (Figure 1). The Cultural Vibrancy subindex explores, on the one side, the provision of cultural infrastructure (*Cultural Venues & Facilities*) and, on the other, the ability of a city to attract visitors and local cultural participation (*Cultural Participation & Attractiveness*), with the aim of providing a synthetic measure describing how liveable and active the cultural scene is. The second subindex explores the Creative Economy domain, with two dimensions assessing, respectively, the current number and creation of new jobs in the cultural and creative sectors (*Creative & Knowledge-based Jobs*, and *New Jobs in Creative Sectors*) and a third dimension synthesising the ability of the city to produce innovative outputs (*Intellectual Property & Innovation*). The picture provided by the first and second subindices, Cultural Vibrancy and Creative Economy, is completed by the third subindex, *Enabling Environment*, which provides insights on *Human Capital & Education*, *Openness, Tolerance & Trust*, *Local & International Connections* (transport) and *Quality of Governance*. The three subindices are aggregated into the ‘C3 Index’, which provides a synthetic measure that brings together the three underlying concepts.



Figure 1. The Cultural and Creative Cities Monitor conceptual framework

2.2. Updated indicators

As anticipated, this year’s release of the monitor is to be regarded as an update in terms of data and methodology, as no substantial changes have been made from a conceptual perspective. The main aims were to maintain consistency, while making sure that the data provided are as reliable as possible.

To this end, as shown in Table 1, the conceptual framework remains mostly unchanged, with an exception made for two indicators: community design application and average appearance in university rankings. The first has been dropped in favour of an indicator on gross expenditure in research and development, as the original indicator has been discontinued by Eurostat. The indicator on average appearance in university rankings has been dropped because it is based on four different rankings (the CWTS Leiden Ranking, QS World University Rankings, Shanghai Ranking and Times Higher Education), adopting different methodologies and rationales to define which universities are included.

In addition, although indicators on satisfaction with cultural facilities (Dimension 1.2, Cultural Participation & Attractiveness), *Tolerance of foreigners*, *Integration of foreigners* and *People trust* (Dimension 3.2, Openness,

Tolerance & Trust)⁽⁴⁾ have been retained in the framework as an indication of their conceptual relevance, all three indicators have been temporarily excluded from this edition because of low coverage and lack of data from 2017 and 2019 (< 30 %, last data available from 2015).

Table 1. Updated conceptual framework

Sub-index	Dimension	Indicator
1. Cultural Vibrancy	1.1 Cultural Venues & Facilities	Sights & landmarks
		Museums & art galleries
		Cinemas
		Concert & music halls
		Theatres
	1.2 Cultural Participation & Attractiveness	Tourist overnight stays
		Museum visitors
		Cinema attendance
		<i>Satisfaction with cultural facilities</i>
2. Creative Economy	2.1 Creative & Knowledge-based Jobs	Jobs in arts, culture & entertainment
		Jobs in media & communication
		Jobs in other creative sectors
	2.2 Intellectual Property & Innovation	ICT patent applications
		Gross expenditure in research & development
	2.3 New Jobs in Creative Sectors	Jobs in new arts, culture & entertainment enterprises
		Jobs in new media & communication enterprises
		Jobs in new enterprises in other creative sectors
3. Enabling Environment	3.1 Human Capital & Education	Graduates in arts & humanities
		Graduates in ICT
	3.2 Openness, Tolerance & Trust	Foreign graduates
		Foreign-born population
		<i>Tolerance of foreigners</i>
		<i>Integration of foreigners</i>
		<i>People trust</i>
	3.3 Local & International Connections	Passenger flights
		Potential road accessibility
		Direct trains to other cities
	3.4 Quality of Governance	Quality of governance

NB: Data for indicators shaded dark grey were not available for this update and therefore these indicators have been omitted.

⁽⁴⁾ These indicators are from the same source: Flash Eurobarometer 366 survey – 'Quality of Life in European Cities'.
https://data.europa.eu/data/datasets/s1035_366?locale=en

2.3. Data sources

As shown in Figure 2, most data (approximately 66 %) come from official sources, and data on essential indicators also come from other sources, including the European Tertiary Education Registry (ETER) and the University of Gothenburg. In addition, all indicators in Dimension 1.1 ‘Cultural venues and facilities’ come from web sources: information from both OpenStreetMap and TripAdvisor contribute to the construction of this dimension, as detailed in Annex 3.

While the approach to data gathering and analysis for the current update of the monitor has remained largely unchanged since the 2019 edition, the shock imposed by the COVID-19 pandemic has had a relevant impact on the availability and timeliness of data. At the time of writing, official data covering both 2020 and 2021 were available for only a small sample of the 196 cities included in this analysis and for a reduced number of indicators. While the two previous editions of the monitor faced the same issue (i.e. lack of data for the exact reference year), this was addressed by adopting a ‘last observation carried forward’ approach: for instance, if data for 2019 were not available, data for 2018 or 2017 were used to replace the missing values. This was considered unfeasible for the 2023 update: data from 2021 would need to be interpreted very differently from data from 2020, which in turn would portray a wildly different picture from 2019 data. Moreover, data from non-official sources might not yet reflect the changes that have happened since the onset of the COVID-19 pandemic. For these reasons, it was decided that the 2023 update of the monitor should be used as an opportunity to revise part of the methodology, the data imputation approach in particular, with the aim of improving time-wise comparability in the process.

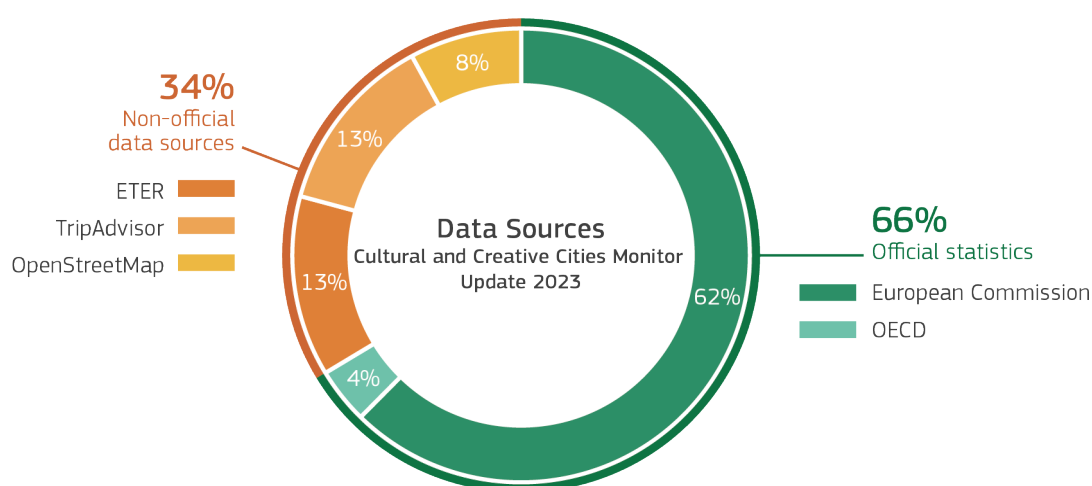


Figure 2. Data sources for the indicators included in the 2023 update of the monitor

NB: European Commission data include data from Eurostat. OECD, Organisation for Economic Co-operation and Development.

2.4. City selection

The 2023 update of the monitor includes 196 European cities with more than 50 000 inhabitants. This threshold was set because of the availability of data in Urban Audit datasets. As in the previous releases, cities are included according to three main selection criteria, namely:

1. cities that have been or will be European Capitals of Culture up to 2028, or that have been shortlisted to become a European Capital of Culture up to 2028;
2. UNESCO Creative Cities;
3. cities hosting at least two regular international cultural festivals.

Since the 2019 edition, 22 new cities have made their way into the monitor (Table 2 shows the full list of new cities included in the ranking), although the overall number of cities analysed has increased by only six. This is because a number of cities are no longer analysed.

- Fourteen cities from the United Kingdom are no longer included in the monitor following the departure of the United Kingdom from the European Union and the resulting discontinuation of its official data provision ⁽⁵⁾.
- Following the redefinition of the Eurostat Urban Audit administrative boundaries in 2021, the city of Terrassa (Spain) is within the boundaries attributed to the 'city' area of Barcelona. The same is true for San Cristóbal de La Laguna, which is among the newly selected cities for the 2023 update but is within the Urban Audit administrative boundaries of Santa Cruz de Tenerife ⁽⁶⁾.
- The city of Limassol (Cyprus) is no longer part of the Eurostat Urban Audit database ⁽⁷⁾.

Table 2. New cities included in the 2023 update of the monitor listed by country

Country	City
AT	Salzburg
BG	Stara Zagora
CZ	České Budějovice
DE	Chemnitz
ES	Valladolid
ES	Santa Cruz de Tenerife
FI	Oulu
FI	Kuopio
FR	Rouen
HU	Miskolc
HU	Kecskemét
IT	Bergamo
IT	Catania
IT	Como
IT	Modena
IT	Pordenone
NL	Tilburg
PL	Gdynia
PT	Ponta Delgada
PT	Aveiro
PT	Viseu
SK	Trenčín

⁽⁵⁾ Glasgow and Liverpool (European Capitals of Culture); Bradford, Bristol, Dundee, Edinburgh, Manchester, Norwich, Nottingham and York (UNESCO Creative Cities); Birmingham, Brighton and Hove, Leeds and London (hosted festivals).

⁽⁶⁾ Geographical reference available at: <https://ec.europa.eu/eurostat/web/gisco/geodata/reference-data/administrative-units-statistical-units/urban-audit>.

⁽⁷⁾ More details and definitions are provided by Eurostat at: <https://ec.europa.eu/eurostat/web/cities/data/database>.

Table 3 provides an overview by category of the number of cities included in the two previous editions and in the 2023 update of the monitor.

Table 3. Cities included in the 2023 update of the monitor by selection criterion

Criterion	2017 count	2019 count	2023 count
European Capital of Culture	92	98	102
UNESCO Creative City	22	33	29
Festivals hosted	54	59	65
Total	168	190	196

The decrease in the number of UNESCO Creative Cities can be attributed to the absence of cities from the United Kingdom in the last edition. Figure 3 provides an overview of all of the cities in the 2023 update of the monitor, and shows which of the three criteria resulted in their inclusion.

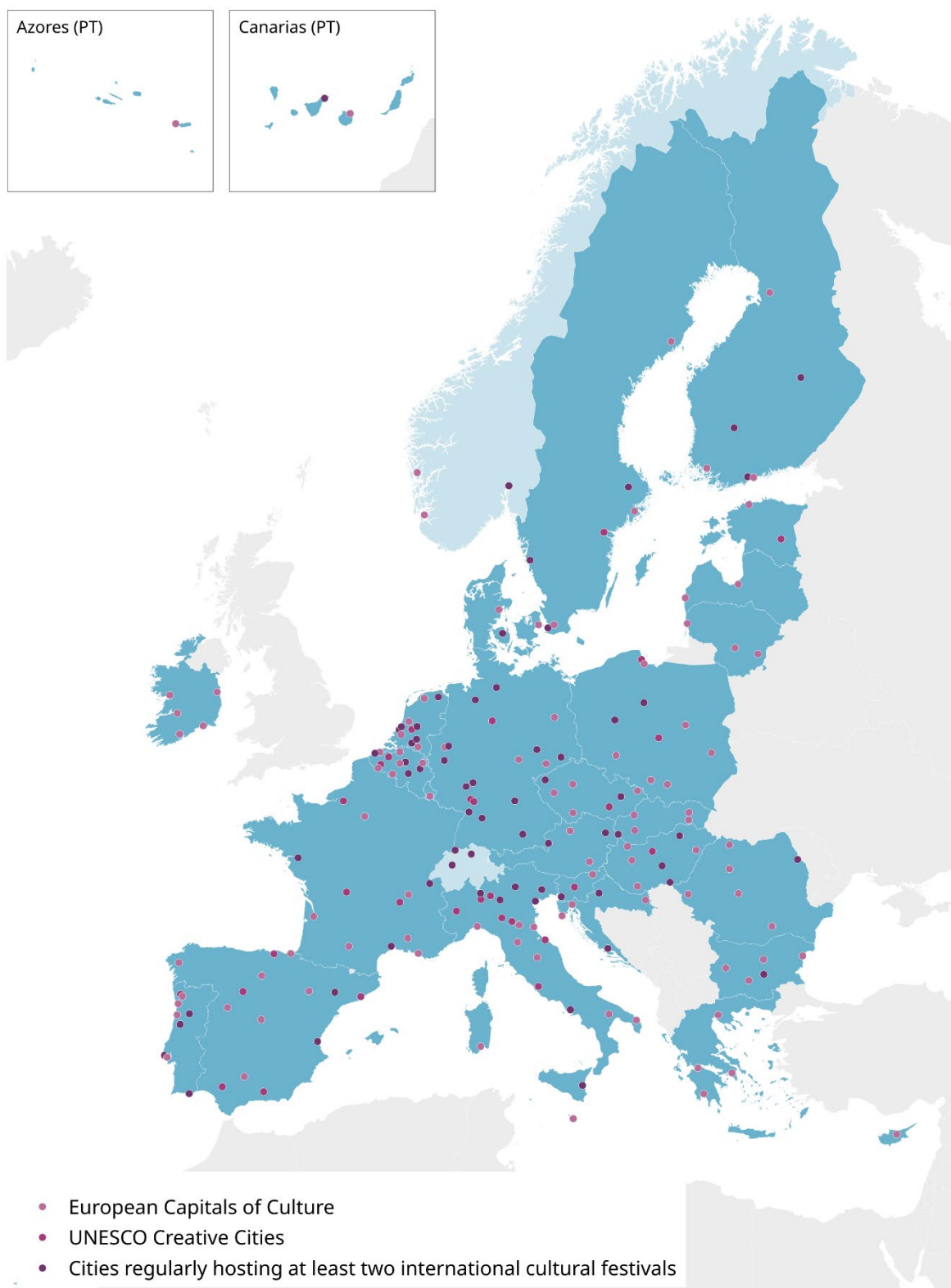


Figure 3. Cities included in the 2023 update of the monitor
 NB: © EuroGeographics for administrative boundaries

2.5. Grouping

To provide meaningful interpretation, the monitor discourages the comparison of widely different cities. In particular, it differentiates cities based on their population size, GDP per capita and employment rate, thus allowing the creation of ‘grouped’ rankings.

Table 4 and Table 5, show the grouping thresholds and group compositions used in the 2023 update of the monitor, respectively. The grouping approach has remained largely unchanged since the last edition, with only the employment rate group ranges being updated, to better reflect the characteristics of the sample of cities analysed. While the employment rate ranges have changed, the methodology used to define the ranges has not: the employment rate groups have again been calculated based on the natural breaks algorithm⁽⁸⁾.

Table 4. Grouping thresholds in the 2023 update of the monitor

Population	Group	GDP per capita	Group	Employment rate	Group
> 1 million	XXL	> 45 000	1	> 78.5 %	1
500 000–1 million	XL	35 000–45 000	2	73.4–78.5 %	2
250 000–500 000	L	27 000–35 000	3	66.2–73.4 %	3
100 000–250 000	M	19 000–27 000	4	54.8–66.2 %	4
50 000–100 000	S	≤ 19 000	5	≤ 54.8 %	5

NB: L, large; M, medium; S, small; XL, extra large; XXL, extra extra large.

Table 5. Groups in the 2023 update of the monitor

Population			GDP per-capita			Employment rate		
Population group	Number of cities per group	Cities per group (%)	GDP per capita group	Number of cities per group	Cities per group (%)	Employment rate group	Number of cities per group	Cities per group (%)
XXL	24	12 %	1	50	26 %	1	53	27 %
XL	33	17 %	2	37	19 %	2	78	40 %
L	44	22 %	3	37	19 %	3	39	20 %
S-M	95	48 %	4	23	12 %	4	21	11 %
			5	49	25 %	5	5	3 %

NB: L, large; M, medium; S, small; XL, extra large; XXL, extra extra large.

⁽⁸⁾ As described in the 2019 edition of the monitor, based on Jenks' (1967) definition, ‘natural breaks ... is a data clustering method designed to determine the best arrangement of values into different classes. This is done by seeking to minimise each class's average deviation from the class mean, while maximising each class's deviation from the means of the other groups. In other words, the method seeks to reduce the variance within classes and maximise the variance between classes’ (Montalto et al., 2019, Annex B, p. 3).

2.6. Data treatment

Most of the indicators (listed in Table 1) represent total numbers of things such as museums, jobs or graduates. These are hard to compare across cities of different sizes, even within the same size group. For this reason, these indicators are expressed per 1 000 city population for the analysis.

Only four of the indicators in our analysis are not divided by the city population: the number of ICT patent applications is divided by the population of the corresponding NUTS (nomenclature of territorial units for statistics) region because these data are not collected at the city level; the gross expenditure in research and development is given as a share of total spending; the foreign-born population is already expressed as a share of the total population; and the quality of governance index is used as is.

2.7. Missing data estimation

Except for the variables in the local and international connections dimension, the city-level datasets used in the monitor, and described in Section 2.1, are incomplete, with missing values for years and/or cities within our sample. Incomplete datasets, in which missing values are either ignored or excluded from analysis, could lead to biased results (e.g. lack of comparability) or to extremely limited final coverage. For this reason, for the datasets used in the monitor, missing values have been filled in using statistical techniques, as described below, in a process known as imputation.

The aim of this process was to fill missing values with values that are reasonable in a broad sense, relying on information from other cities, years or variables to define such reasonable values. Whenever possible, a dataset larger than the one ultimately used for the final indicator was employed for the purposes of imputation. This has increased the amount of information that can be drawn on. For example, the raw dataset for tourist overnight stays included over 800 cities with observations starting from 1990.

The imputation method used involves estimating a time trend for a given indicator using data on many cities and applying this time trend to cities for which some data points are missing. This step assumes that any differences between cities are constant, an assumption that is often not warranted for the indicators used. For this reason, indicators were transformed before imputation in such a way that the assumption could more reasonably be made. These transformations were reversed after imputation was complete.

Our process can be described in three steps: (1) the indicators were transformed to fit the imputation method, (2) time trends were estimated for each indicator and (3) the time trends were applied to missing data points and the transformations reversed.

2.7.1. Transformation for imputation

The second column of Table 6 describes the transformations undertaken for each indicator before imputation. Each involved a judgement call along the lines outlined above.

The most common transformation applied is the natural logarithm, which is helpful where distributions are heavily skewed. A logit transformation is applied for variables that are expressed as shares or percentages. This ensures that small differences between very small values (close to 0 %) or between very large values (close to 100 %) are properly weighted in the imputation.

For the quality of governance index, the mean of the regional group was subtracted from the variable. This variable contains many missing values and varies significantly across regions. The regional de-meaning is meant to make values comparable, so that a common time trend can be estimated for all cities for which data are available.

2.7.2. Trend estimation

For each (transformed) indicator, time trends were then estimated that describe how the indicator changed over time on average. When enough non-missing data points were available for an indicator, time trends were estimated separately by groups of cities. These groups were based on either city population or GDP per capita, depending on which of these measures was judged to be more likely to correlate with differences in the development of an indicator over time. The third column of Table 6 shows the number of these groups and the variable by which the grouping was performed.

Table 6. Details on imputation method for each variable imputed

Indicator	Transformation for imputation	Groups
Sights & landmarks	Log	Three, by mean city population
Museums & art galleries	Log	Three, by mean city population
Cinemas	Log	Three,, by mean city population
Concert & music halls	Log	Three, by mean city population
Theatres	Log	Three, by mean city population
Tourist overnight stays	None	Three, by mean city population
Museum visitors	None	Three, by mean city population
Cinema attendance	Log	Three, by mean city population
Jobs in arts, culture & entertainment	None	Three, by mean city population
Jobs in media & communication	None	Three, by mean city population
Jobs in other creative sectors	None	Three, by mean city population
ICT patent applications	Log	Three, by mean city population
Gross expenditure in research & development	Logit	Three, by GDP per capita
Jobs in new arts, culture & entertainment enterprises	Log	Three, by GDP per capita
Jobs in new media & communication enterprises	Log	Three, by GDP per capita
Jobs in new enterprises in other creative sectors	Log	Three, by GDP per capita
Graduates in arts & humanities	Log	None
Graduates in ICT	Log	None
Foreign graduates	Log	None
Foreign-born population	Logit	Three, by mean city population
Quality of governance	De-mean by region	None

Each time trend was estimated using a regression analysis that seeks to explain the available data points (i.e. the values of an indicator for a set of cities), as a function of time. To allow for smooth but non-linear time trends, a set of cubic splines was used in the regressions ⁽⁹⁾. The estimated time trends were used for imputation only within the period for which data were available, so as not to rely on so-called out-of-sample predictions ⁽¹⁰⁾.

2.7.3. Imputation

Imputation for missing values was carried out at four different levels, depending on which related data were available.

1. **Interpolation.** If data points were available for the same city both before and after a missing value, the value was filled by linear interpolation (i.e. a linear increase or decrease is assumed between the nearest previous value of the transformed indicator and the nearest following value).
2. **Extrapolation.** If data points were available for the same city only either before or after a missing value, then the estimated trend was used along with the nearest existing data point to impute. To extrapolate values for a given city, the trend estimated for the group to which the city belongs was shifted up or down to fit the nearest existing data point for the city and then followed to the missing year to determine the imputed value.
3. **Trend only.** If no data points were available for a given city, then all values for that city were set equal to the estimated trend for the grouping to which the city belongs ⁽¹¹⁾.
4. **Closest imputed.** For the three indicators that were obtained from TripAdvisor, Sights & landmarks, Museums & art galleries, Concert & music halls), no information was available from before 2017. The missing values for 2015 and 2017 were filled using the closest observed or imputed values (i.e. the values for 2017).

Finally, the transformations used for imputation were reversed. For instance, the exponential function was applied to the log number of cinemas per capita and regional means were added back to the imputed values of the quality of governance index.

2.8. Normalisation, weighting and aggregation

The approaches to normalisation, weighting and aggregation have remained substantially unchanged since the 2019 edition of the monitor ⁽¹²⁾. Namely, the data have been normalised on a scale of 0–100, where 0 is the lowest value and 100 is the highest value for each indicator, thus ensuring consistency among indicators in the framework.

The weighting values used, defined with input from experts during the development of the first edition of the monitor, are intended to highlight the particular relevance of the cultural vibrancy and creative economy subindices (both 40 %) and the complementarity of the third subindex, Enabling Environment (20 %).

At the dimension level, equal weighting has been attributed to the dimensions of the cultural vibrancy subindex. In the Creative Economy subindex, the dimensions related to jobs are attributed a 40 % weighting, while the dimension on *Intellectual Property & Innovation* has a lower weighting (20 %). Different weightings have also been defined for the third subindex, Enabling Environment, with *Human Capital & Education and Openness, tolerance & Trust* being attributed a weighting of 40 % each, and *Local & International Connections* and *Quality of Governance* being attributed weightings, respectively, of 15 % and 5 %.

⁽⁹⁾ Cubic splines are a mathematical tool commonly used in data analysis to estimate smooth and continuous curves. They are particularly useful when dealing with complex data patterns that may contain non-linear relationships.

⁽¹⁰⁾ For example, if no data were available for a given indicator before 2016, then a trend estimated based on these data is used to impute missing values for 2016, but never for 2015. In three cases, the subsequent analysis required data points outside the range given by the data. In these cases, another approach was used that did not rely on the trend (see point 4 in section 2.7.3).

⁽¹¹⁾ This process is very similar to taking the average level of the cities in the same group.

⁽¹²⁾ More details are available in *The Cultural and Creative Cities Monitor – 2019 edition* (Montalto et al., 2019, Annex A).

2.9. Comparability

2.9.1. Comparability over time

To ensure comparability over time, data in the 2019 edition were recalculated to cover observations pertaining to the 2019 edition's reference period⁽¹³⁾.

In the 2023 update, while aiming to maintain consistency with the 2017 and 2019 editions, a different approach was adopted to improve the comparability over time, with a clear reference to a 1-year-only observation (or imputation). While the 2017 and 2019 editions of the monitor used data for a range of years to populate each indicator, the 2023 update takes into account only data in the reference year, observed or imputed. This approach means that, on the one hand, more missing data will be present in the initial dataset, as the criteria used to define which observations to use has changed (as outlined in section 2.3), while, on the other hand, it allows more meaningful comparisons to be made over time.

2.9.2. Comparability between cities

From a spatial perspective, the definition of cities remains unchanged from the 2019 edition, relying on the boundaries provided for cities in the Urban Audit initiative by Eurostat. In particular, the current update of the monitor relies on the 2021 Urban Audit definition of cities⁽¹⁴⁾, which aims to ensure comparability between urban areas. Compared with the Urban Audit areas used in the 2019 edition of the monitor, some notable changes have taken place, including the redefinition of borders for the cities of Paris (France) and Milan and Naples (Italy) to include areas surrounding the main local administrative units identifying the administrative 'communes'⁽¹⁵⁾. In addition, some cities have been incorporated within the boundaries of neighbouring cities. This is the case for the Spanish cities of Terrassa, incorporated within the Urban Audit boundaries of Barcelona, and San Cristóbal de La Laguna, incorporated within the Urban Audit boundaries of Santa Cruz de Tenerife.

These changes do not affect comparability among the sample of cities analysed in the 2023 update. However, data from the 2023 update of the monitor are not comparable with data from the two previous editions.

⁽¹³⁾ More details are available in *The Cultural and Creative Cities Monitor – 2019 edition* (Montalto et al., 2019, Annex A).

⁽¹⁴⁾ <https://ec.europa.eu/eurostat/web/gisco/geodata/reference-data/administrative-units-statistical-units/urban-audit>.

⁽¹⁵⁾ More details on the definition of 'cities' are available at Eurostat ([https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Territorial typologies manual - cities, commuting zones and functional urban areas#Links to other spatial concepts.2Ftypologies](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Territorial_typologies_manual_-_cities,_commuting_zones_and_functional_urban_areas#Links_to_other_spatial_concepts.2Ftypologies)) and in the Urban Europe report (Eurostat, 2016).

3. Statistical assessment

3.1. Data availability

Table 7 provides an overview of the imputation method for dealing with missing data adopted at indicator level.

Table 7. Observed and imputed indicator data for 2019 data points

Sub-index	Dimension	Indicator	Observed	Interpolated	Extrapolated	Trend
Cultural Vibrancy	Cultural Venues & Facilities	Sights & landmarks	89%	0%	11%	0%
		Museums & art galleries	89%	0%	11%	0%
		Cinemas	97%	0%	1%	2%
		Concert & music halls	88%	0%	10%	2%
		Theatres	98%	0%	0%	2%
	Cultural Participation & Attractiveness	Tourist overnight stays	64%	1%	29%	7%
		Museum visitors	39%	9%	44%	8%
		Cinema attendance	44%	3%	42%	11%
		Satisfaction with cultural facilities				
Creative Economy	Creative & Knowledge-based Jobs	Jobs in arts, culture & entertainment	67%	0%	22%	11%
		Jobs in media & communication	67%	0%	22%	11%
		Jobs in other creative sectors	67%	0%	22%	11%
	Intellectual Property & Innovation	ICT patent applications	68%	0%	17%	14%
		Gross expenditure in research & development	87%	1%	1%	12%
	New Jobs in Creative Sectors	Jobs in new arts, culture & entertainment enterprises	77%	1%	0%	23%
		Jobs in new media & communication enterprises	77%	1%	0%	23%
		Jobs in new enterprises in other creative sectors	77%	0%	0%	23%
Enabling Environment	Human Capital & Education	Graduates in arts & humanities	67%	0%	19%	14%
		Graduates in ICT	64%	0%	19%	16%
	Openness, Tolerance & Trust	Foreign graduates	78%	0%	12%	10%
		Foreign-born population	10%	6%	33%	51%
		Tolerance of foreigners				
		Integration of foreigners				
		People trust				
	Local & International Connections	Passenger flights	100%	0%	0%	0%
		Potential road accessibility	100%	0%	0%	0%
		Direct trains to other cities	100%	0%	0%	0%
	Quality of Governance	Quality of governance	0%	77%	0%	23%

NB: The quality of governance indicator has observed values for 2010, 2013, 2017 and 2021. Therefore, no observed values exist for 2019. However, observations for 2019 are imputed. Data for indicators shaded dark grey have not been included in this update of the monitor (latest observed values are too old).

3.2. Statistical coherence

The assessment of statistical coherence consists of a multilevel analysis of the correlations of variables, and a comparison of rankings at the C3 Index level with their constituent subindices and dimensions.

3.2.1. Correlation analysis

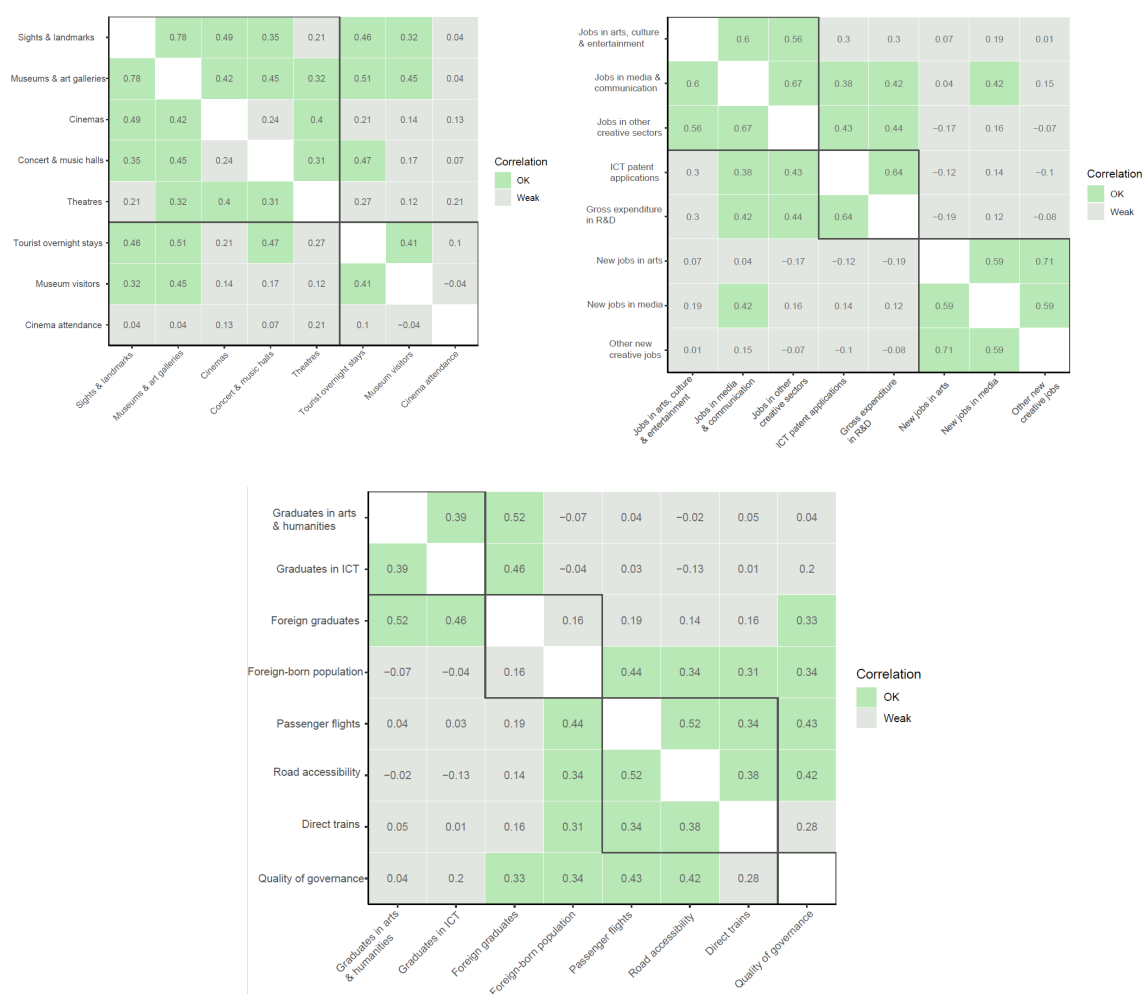
As suggested by the Joint Research Centre's Competence Centre on Composite Indicators and Scoreboards (JRC-COIN), for the statistical assessment of composite indicators, the statistical coherence of an index should be considered a useful tool for evaluating the soundness of an index, despite it not being sufficient alone (Caperna, 2022). The main tool used for this assessment is correlation analysis, because it helps to assess the extent to which the observed data are in line with the conceptual framework. Within each level of the index, there should ideally be positive significant correlations (JRC-COIN suggests ≥ 0.30). This makes it more likely that the overall index scores will adequately reflect the values of the underlying indicators. The framework should avoid redundancy, which can be identified by very high correlations (≥ 0.92). This is because, if two indicators are collinear, then this may result in the double counting (and thus overweighting) of the same phenomenon.

In what follows, we report the correlations between indicators in the same subindex, between indicators and their relative aggregates, and finally between dimensions, subindices and the C3 Index. The analysis results and figures presented in this section (Section 3.2) were derived from the COINr package developed by JRC-COIN (Becker et al., 2022).

3.2.1.1. *Correlation analysis between indicators of the same group*

The first step in the analysis of internal coherence is based on the correlation between indicators included in the same aggregation groups. Table 8 shows the correlation coefficients between indicators within the same subindex. Boxes within each subindex identify indicators grouped into respective dimensions. The correlation between indicators of the same dimension is the most important aspect to observe here. We can see that the vast majority of the indicators are well correlated with the others within their dimensions. There are two exceptions. First, the indicator of cinema attendance in the Cultural Participation and Attractiveness dimension shows a correlation with the other indicators of this dimension that is close to zero or slightly negative; this may suggest that this indicator carries an underlying message that does not have much in common with the others. The main risk in this kind of situation is under-representing the uncorrelated indicator in the aggregates (dimension, subindex, index). Fortunately, this indicator shows a satisfactory correlation with all the higher-level aggregates (see Table 9). The second case of low correlation is between the two indicators of the Openness, Tolerance and Trust dimension (a correlation of 0.16 between foreign graduates and foreign-born population). This implies that these two indicators will conflict a little when aggregated, but not in a strong way, considering that the correlation is not negative. In observing the relation of the two indicators with the openness, tolerance and trust dimension, the indicator foreign-born population shows more relevance and similarity with the aggregate (correlation of 0.93 v 0.52; see Table 9), suggesting that it may be more represented than the foreign graduates indicator in the following steps of aggregation. Nevertheless, the correlations of the two indicators with the subindex and the index prove to be similar and sufficiently strong, confirming a balanced representation of the two indicators in the index.

Table 8. Correlations between indicators in the same subindex



NB: Numbers represent Pearson correlation coefficients. Good correlations (i.e. Pearson correlation coefficients of > 0.30 and < 0.92) are shaded light green. Correlations with low values (here < 0.30) are shaded grey.

Further observation of Table 9 shows that almost all indicators have a good correlation with their relative aggregates. In a few cases, the correlations with the C3 Index seem a bit small but not in a drastic way. Specific mention is needed for Jobs in new arts, culture, and entertainment, which has a correlation close to zero with the index. This indicator is not well represented in the final aggregate, which does not mean that it has no impact. This effect is mainly due to the relation of the New jobs dimension with the other dimensions of the subindex, which is commented on below.

Table 9. Correlations between indicators and their aggregates

	Dimension	Sub-index	Index
Sights & landmarks	0.82	0.75	0.27
Museums & art galleries	0.85	0.78	0.39
Cinemas	0.68	0.59	0.36
Concert & music halls	0.62	0.61	0.41
Theatres	0.61	0.58	0.51
Tourist overnight stays	0.75	0.72	0.54
Museum visitors	0.26	0.37	0.36
Cinema attendance	0.73	0.41	0.35
Jobs in arts	0.8	0.64	0.59
Jobs in media	0.89	0.82	0.66
Other creative jobs	0.88	0.69	0.61
ICT patent applications	0.92	0.58	0.53
Expenditure in R&D	0.89	0.57	0.54
New arts jobs	0.81	0.32	0.04
New jobs in media	0.89	0.67	0.33
Other new creative jobs	0.88	0.42	0.22
Graduates in arts	0.63	0.36	0.26
Graduates in ICT	0.96	0.51	0.23
Foreign graduates	0.52	0.64	0.41
Foreign-born population	0.93	0.69	0.45
Passenger flights	0.83	0.59	0.36
Road accessibility	0.72	0.43	0.29
Direct trains	0.76	0.47	0.37
Quality of governance	1	0.63	0.47

Correlation

High

OK

Weak

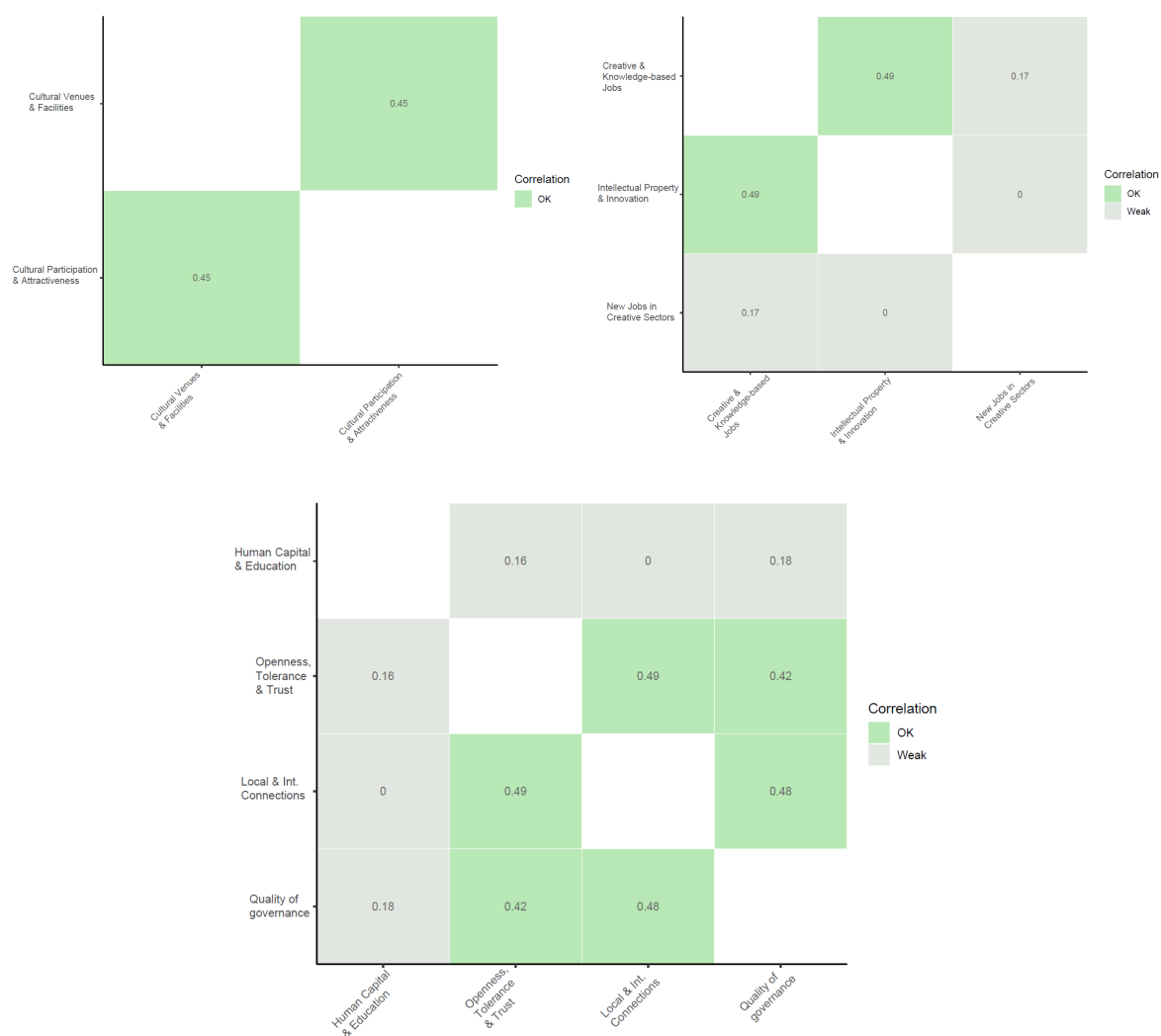
NB: Numbers represent Pearson correlation coefficients. Good correlations (i.e. Pearson correlation coefficients > 0.30 and < 0.92) are shaded light green. Correlations with low values (here < 0.30) are shaded grey. Correlations suggesting a risk of redundancy (here > 0.91) are shaded darker green.

3.2.1.2. Correlation analysis between aggregates and with the index

Correlation analysis at the dimension level

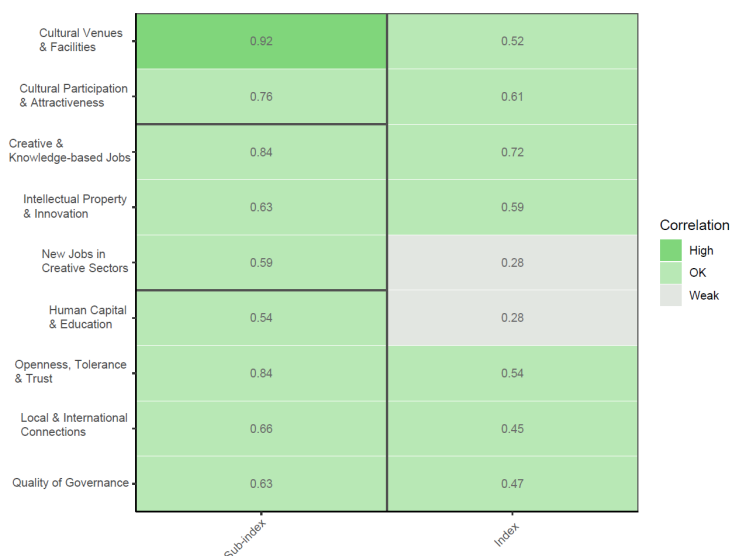
The analysis of correlations between the aggregates, especially in the top levels, is important because it reflects the consistency of the defined concepts. Among the nine dimensions, only two show low correlations with all the other dimensions of their subindex: New jobs in the Creative Sector and Human Capital & Education (Table 10). In both cases, the dimensions, despite the clear internal correlation, are not aligned with the other elements of their subindices. This is because the indicators of these two dimensions are correlated within the dimensions, but have mostly negative correlations with the other elements of the subindex (see Table 8). In this phase, there is no reason to consider any modification of these elements of the index. This is because the structure of the index comes from previous editions, where the coherence with the rest of the framework was stronger, and the consistency of the tool requires it to remain as similar as possible in structure to the previous editions. Moreover, Table 11 shows how these two dimensions are penalised because they are the least related of their groups with the higher aggregates (subindex and index), but the penalisation observed is not sufficient to identify the two dimensions as irrelevant or under-represented (correlation with the index equal to 0.28). Future editions of the index may need to pay specific attention to these elements, at both the dimension and indicator levels. In general, the correlation of dimensions with the C3 Index is balanced (0.45 – 0.72, excluding those already discussed).

Table 10. Correlation analysis between dimensions



NB: Numbers represent Pearson correlation coefficients. Good correlations (i.e. Pearson correlation coefficients > 0.30 and < 0.92) are shaded light green. Correlations with low values (here < 0.30) are shaded grey.

Table 11. Correlation of dimensions with subindices and index

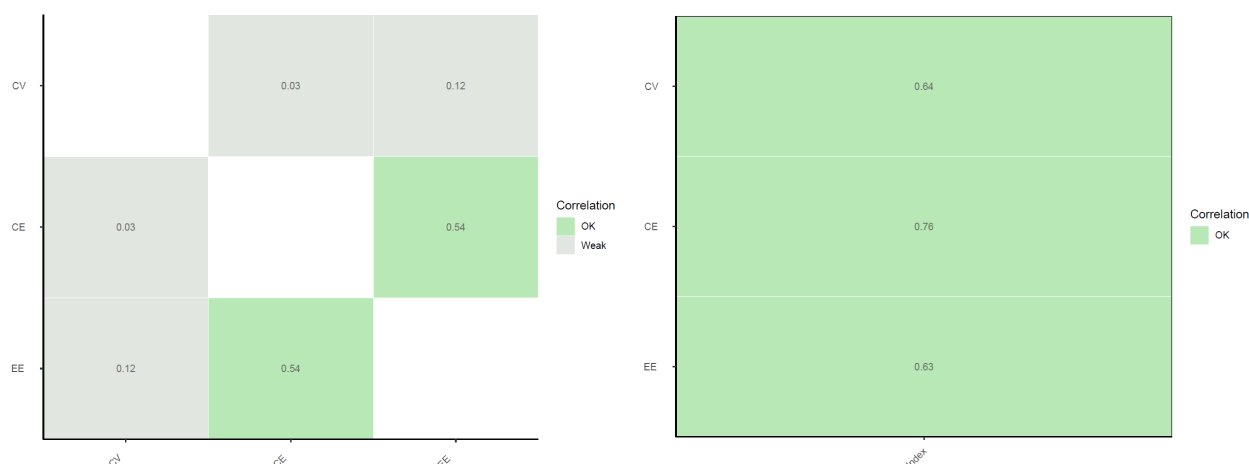


NB: Numbers represent the Pearson correlation coefficients. Good correlations (i.e. Pearson correlation coefficients > 0.30 and < 0.92) are shaded light green. Correlations with low values (here < 0.30) are shaded grey. Correlations suggesting a risk of redundancy (here > 0.91) are shaded darker green.

Correlation analysis of subindices

The highest aggregation steps, between subindices and from subindex to index, display a statistical relation that requires attention in the interpretation. High statistical reliability among the main components can be the result of redundancy of information. As shown in Table 12, overall, the subindices of Creative Economy and Enabling Environment are positively related, whereas Cultural Vibrancy seems to share a very limited proportion of its variance with them. This is not reflected in the correlation of the three subindices with the C3 Index, as they are all correlated in a strong and balanced way. Nevertheless, this lack of correlation may represent a limit to the robustness of the index.

Table 12. Correlations between the subindices and with the index



NB: Numbers represent Pearson correlation coefficients. Good correlations (i.e. Pearson correlation coefficients > 0.30 and < 0.92) are shaded light green. Correlations with low values (here < 0.30) are shaded grey. CE, Creative Economy; CV, Cultural Vibrancy; EE, Enabling Environment.

3.2.2. Principal component analysis of the C3 Index

In a further step in the analysis of statistical coherence, we performed a principal component analysis (PCA). The aim of PCA is to assess to what extent the conceptual framework is confirmed by statistical approaches. The objective is to observe only one principal component that explains the large majority of the total variance (approximately 70%), which would suggest the presence of a common, unidimensional phenomenon underlying the subindices.

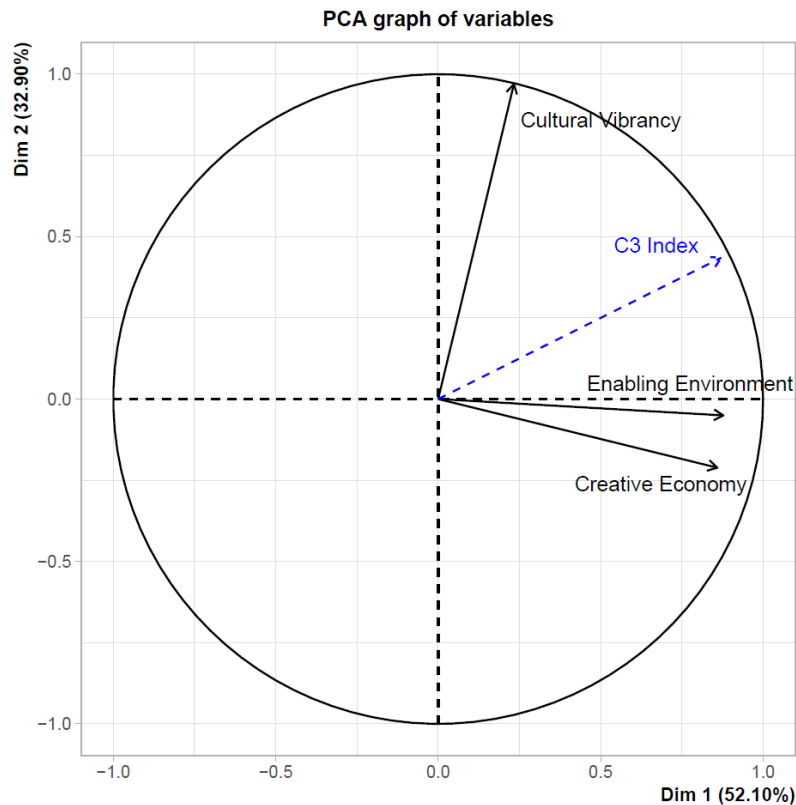


Figure 4. Factor map of the three subindices and comparison with the overall C3 Index

As shown in Figure 4, the three subindices cannot be largely explained by a single statistical dimension because the first principal component summarises only 52.1 % of the total variance. The second principal component is influential because it accounts for 32.9 % of the total variance and describes mainly one subindex. This result depends on the correlation structure: the PCA shows a higher level of agreement between the subindices of Enabling Environment and Creative Economy, while Cultural Vibrancy seems to describe a different concept, not a conflicting concept, just an unrelated one. The distance between the behaviour of Cultural Vibrancy and the other subindices was partly observed in the previous edition of the C3 Index, but in this update it is stronger. Future editions of the index may benefit from a general revision and update of this PCA, to gain a better understanding of the effects of future cultural trends on the structure of the index.

3.2.3. Impact of the modelling assumptions on the C3 Index results

An informative step in the analysis of a composite indicator is the assessment of the effect of different modelling assumptions among reasonable alternatives. Some of these choices may be based on experts' opinion or technical considerations, driven by statistical analysis or by criteria related to the communication of the index. This analysis examines to what extent some of the decisions involved in creating the C3 Index have an impact on the ranking of cities. It is desirable for any composite index to remain relatively unchanged if some of the steps taken to create it are changed. If small changes in methodology have a large impact on the final ranking of an observed unit, then it can be said that the index creates a ranking with a high level of uncertainty for that unit.

The uncertainty analysis here includes changing the procedure used for the treatment of outliers and the perturbation of weighting at the dimension and subindex levels, as outlined in Table 13. These steps were chosen, as there are plausible alternative pathways for each of them in the construction of the index that can be relatively easily investigated. Other assumptions were tested in the previous edition of the index.

Table 13. Alternative assumptions considered in the uncertainty analysis

Assumption	Reference	Alternative
Maximum number of winsorised values before log transformation	5	1–5
Weighting system for subindices and dimensions	Nominal weighting	Perturbed weighting ($\pm 25\%$)

In the following sections, the results of this analysis are briefly shown for every population group. The results suggests that the main source of uncertainty is the perturbation of weighting. This is mainly due to the correlation structure of the index. In particular, the low correlation between the Cultural Vibrancy subindex and the other subindices causes the high influence the weights show. Further details are presented below.

3.2.3.1. Extra extra large cities

As shown in Figure 5, among the extra extra large (XXL) cities, only Barcelona and Budapest have an interval of ranks of larger than 5. This suggests that they are more influenced by the variation of the criterion on winsorisation than others. Without this criterion, the interval of ranks for these cities is 5 or less, but most of the variability still remains because it is related to the perturbation of weighting.

It should be noted that a more restrictive approach (e.g. an interval of > 3 instead of > 5) would highlight some instability at the top ranks. The medians are close to the nominal ranking, and this can be considered proof that the nominal rankings are representative of a plurality of scenarios.

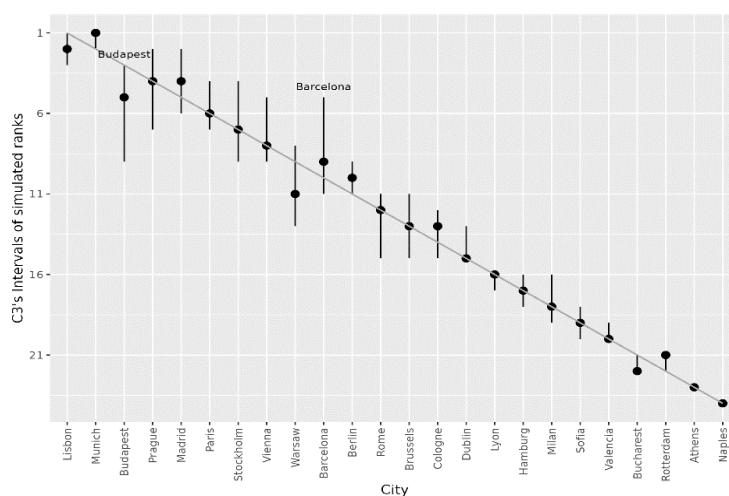


Figure 5. Impact of the modelling assumption for XXL cities

3.2.3.2. Extra large cities

Figure 6 shows that, among the extra large (XL) cities, seven have an interval of ranks of larger than 5, and these cities are close to each other in the ranking: this indicates that their scores are probably similar or unbalanced and so they could easily switch their positions and are influenced by weighting.

Excluding Zagreb, the variability depends on the weighting and is particularly visible among mid- and high-performing cities. The remaining ranks are robust.

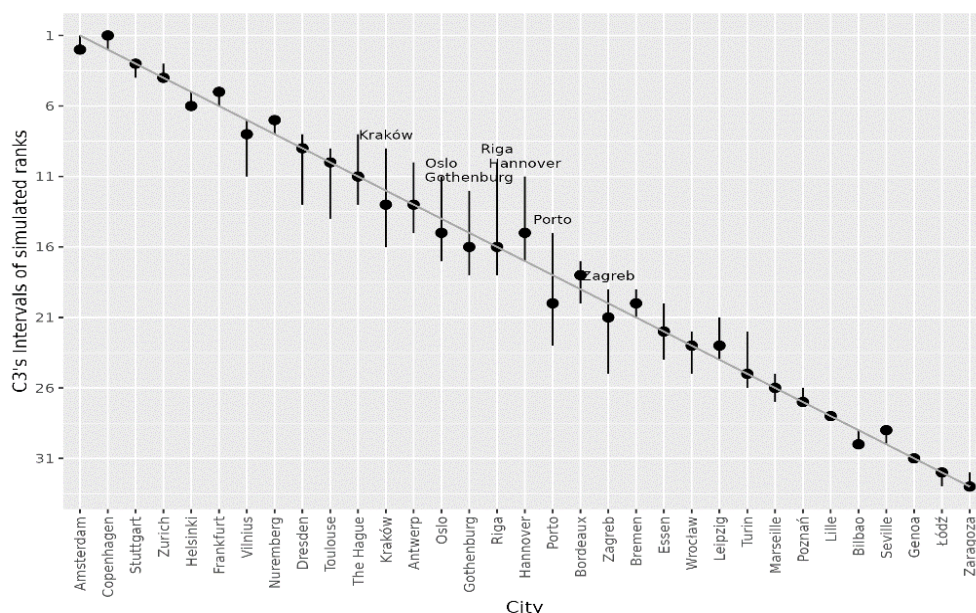


Figure 6. Impact of the modelling assumption for XL cities

3.2.3.3. Large cities

Figure 7 shows that, as for the previous group, the large (L) cities group is very robust, although seven cities show more variability. Also in this case, most variability can be attributed to weighting, although few cities with larger degrees of uncertainty are isolated and their large interval is due to their specific characteristics.

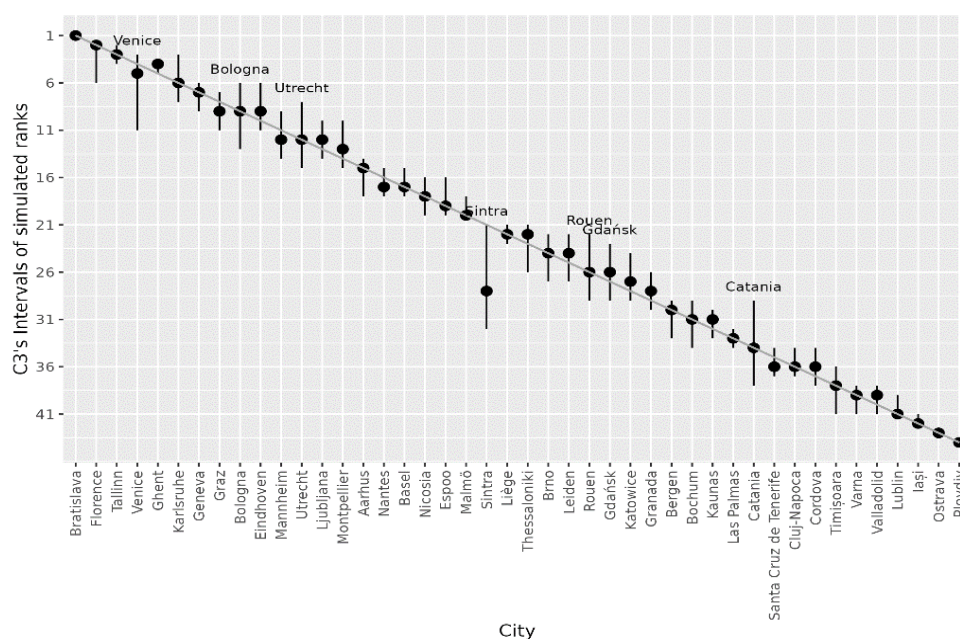


Figure 7. Impact of the modelling assumption for L cities

3.2.3.4. Small and medium cities

S and M cities are usually more specialised and are cultural and creative in different ways. This is also true for the impact of the assumption on winsorisation, which is large for a few cities, as shown in Figure 8.

When accounting for the effects of winsorisation, there is high variability across the sample, with some extremes for some cities. Nevertheless, the main source of variability are the weighting values, thus substantiating the hypothesis that the cities analysed might be cultural and creative in their own specific ways.

The ranking of the middle and low part of the group should not be used for direct inference because it is dependent on the assumptions.

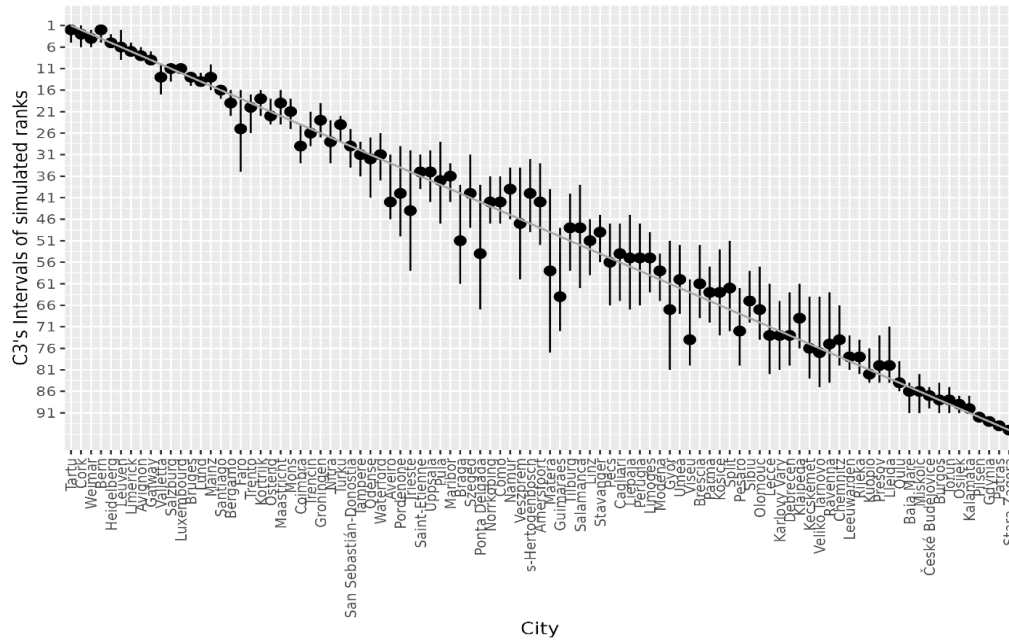


Figure 8. Impact of the modelling assumption for S and M cities

4. Online platform

Since its first edition in 2017, the monitor has relied on an online tool⁽¹⁶⁾ that allows users to explore data for each city, thus allowing meaningful comparisons based on the population, GDP per capita and employment rate groups.

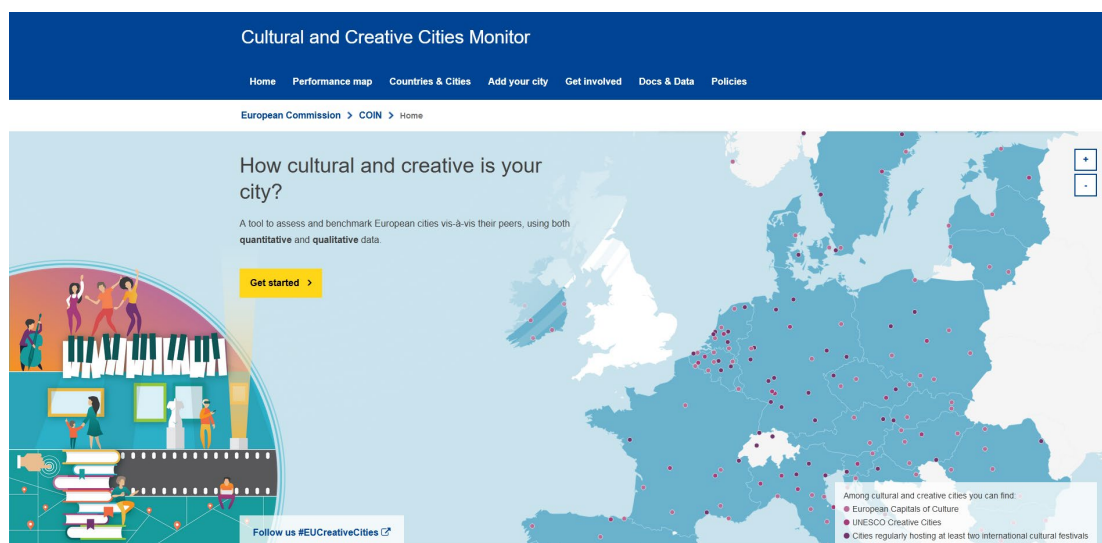


Figure 9. Home page of the monitor's online tool

NB: the figure above is only meant to provide an example of the visualisations. Refer to the online platform for the most updated version.

The dashboard of this update of the monitor (Figure 9) retains most of the functionalities of the previous edition, while including an additional layer of information on the evolution of the performance of the cities at both an aggregate level and an indicator level. The 2023 update of the platform also allows users to explore the evolution of the performance of the cities over time.

The following sections provide a quick tour of the main features of the tool.

4.1. Visually explore European cultural and creative cities

The monitor does not offer one single ranking to assess culture and creativity in European cities. Instead, its online tool provides different rankings depending on the point of observation (Figure 10). For example, it is possible to use the performance map to visualise the top five cities for the subindex Cultural Vibrancy, Creative Economy or Enabling Environment. Moreover, rankings are provided for each of the three groupings: population, GDP per capita and employment rate. Each ranking can be explored for the three years observed: 2015, 2017 and 2019.

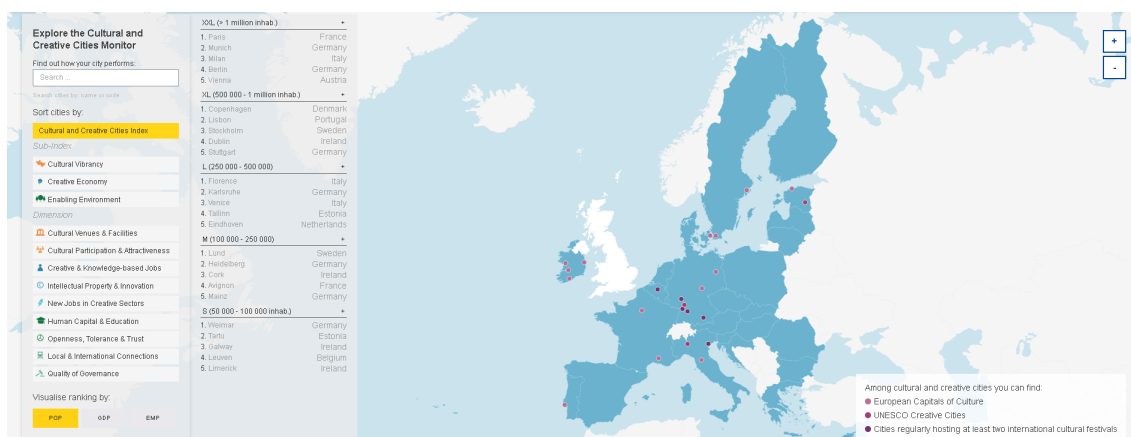


Figure 10. Performance map of the monitor

NB: the figure above is only meant to provide an example of the visualisations. Refer to the online platform for the correct rankings.

⁽¹⁶⁾ The Cultural and Creative Cities Monitor – online tool (<https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor/>).

The 25 cities highlighted on the map are the top five cities appearing in the ranking, capturing cultural infrastructure and participation in each group (Figure 10 shows cities with a similar population size). Each of the group rankings can be explored in full by selecting the ‘+’ icon at the top of each ranking.

4.2. Compare and contrast similar cities

The monitor’s online tool provides 196 city profiles summarising the main aspects captured by the composite indicator using three main interactive visualisations.

The overall performance visualisation (Figure 11) displays the full ranking and the cities’ performances in the C3 Index, in the three subindices or in the nine dimensions.

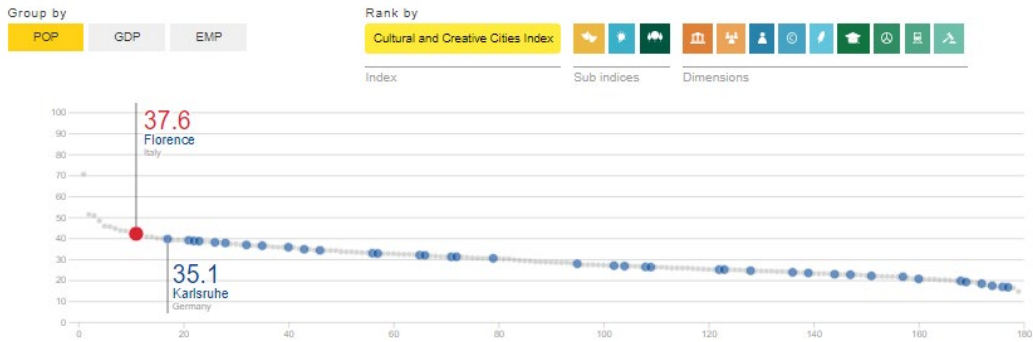


Figure 11. Overall performance: the monitor – 2023 update
NB: the figure above is only meant to provide an example of the visualisations. Refer to the online platform for the correct rankings.

The second visualisation of the city profile (Figure 12) offers a more detailed picture of the city performance, allowing the user to explore how cities perform in each dimension and the underlying indicators.

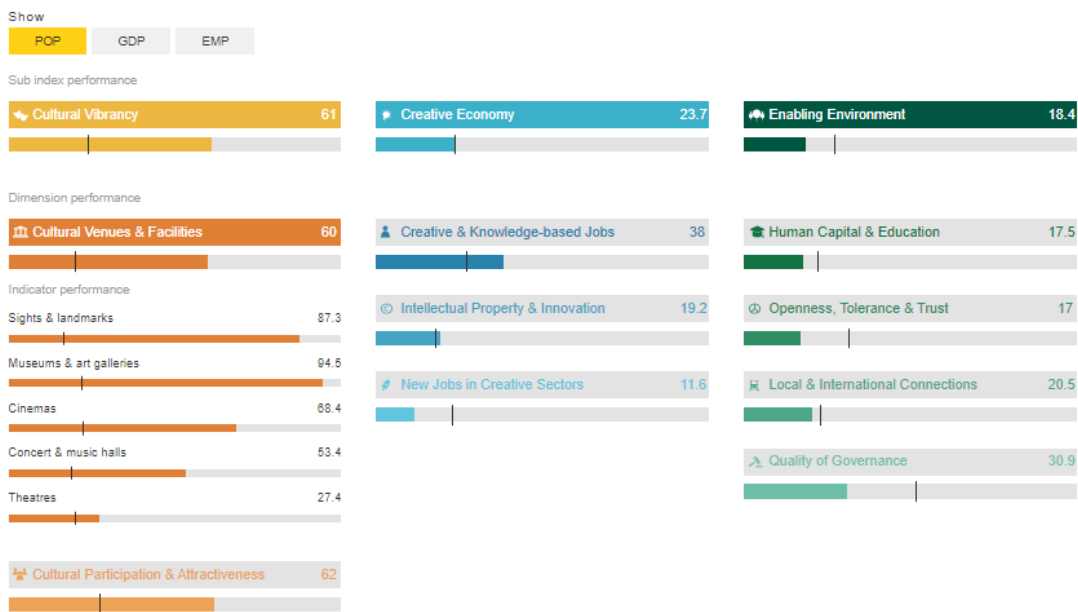


Figure 12. Performance by dimension
NB: the figure above is only meant to provide an example of the visualisations. Refer to the online platform for the correct data.

Figure 13 shows a comparison of the performances of Florence with the average values for other similar cities. As shown in Figure 12, Florence is the best performer among cities, having between 250 000 and 500 000 inhabitants. It performs better than average in cultural vibrancy and its dimensions (*Cultural Venues & Facilities*; *Cultural Participation & Attractiveness*). Despite this, Florence is aligned with the average performance for *Creative Economy* and is below average for *Enabling Environment*. This suggests that even top performers like Florence can improve to reinforce or empower cultural and creative factors in the city.

Search peer cities ...		POP	GDP	EMP
Show		Differences		
	Florence (IT)	Venice (IT) ✕	Nantes (FR) ✕	
Cultural and Creative Cities Index	37.6	34.5 (-3.1)	26.6 (-11.0)	
- 🏛 Cultural Vibrancy	61.0	58.8 (-2.2)	30.4 (-30.6)	
- 🏛 Cultural Venues & Facilities	60.0	55.8 (-4.3)	31.5 (-28.5)	
Sights & landmarks	87.3	100.0 (+12.7)	6.2 (-81.1)	
Museums & art galleries	94.5	100.0 (+5.5)	10.9 (-83.6)	
Cinemas	68.4	26.2 (-42.1)	46.8 (-21.5)	
Concert & music halls	53.4	70.9 (+17.5)	21.1 (-32.4)	
Theatres	27.4	25.9 (-1.5)	49.5 (+22.1)	
+ 🏛 Cultural Participation & Attractiveness	62.0	61.9 (-0.1)	29.3 (-32.7)	
+ 🏛 Creative Economy	23.7	17.7 (-6.0)	22.4 (-1.3)	
+ 🏛 Enabling Environment	18.4	19.5 (+1.1)	27.4 (+8.9)	

Figure 13. Compare my city with...

NB: the figure above is only meant to provide an example of the visualisations. Refer to the online platform for the correct data.

A third visualisation (Figure 13) allows the user to select peer cities and compare each aspect of the framework (overall index, subindices, dimensions and indicators). The performance of several cities can be displayed side by side and differences can be observed, allowing the user to distinguish between cities performing well (black) and less well (grey).

4.3. Follow the project and use the data

A section entitled 'Docs & Data' ⁽¹⁷⁾ provides an overview of the past editions of the monitor, along with publications related to the project. The full dataset used to produce the monitor has also been made available for the 2017 and 2019 editions and the 2023 update.

⁽¹⁷⁾ <https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor/docs-and-data>

5. Conclusions

The 2023 update of the monitor was developed with the aim of building on the monitor's strong framework, while making methodological adjustments to ensure that results are comparable over time. It should be noted that the data and analyses provided in the online tool of the 2023 update are not comparable with those of the 2017 and 2019 editions of the monitor. An effort to maintain comparability across cities is made by relying on Urban Audit boundaries, which are based on the European Commission's definition (¹⁸), which aims to provide a harmonised – and thus comparable – definition of cities. However, Urban Audit boundaries have limitations in defining city areas: in addition to features characterising contiguous dense urban areas, they also take into account administrative boundaries, which are in general not designed with comparability with other cities in mind.

The updated results are available for three reference years, 2015, 2017 and 2019, for a total of 196 European cities, and allow policymakers to compare cities' relative strengths both across cities of similar size and over time.

The revision of the methodology will also simplify data gathering and analysis once updated official data become available. This will allow changes in the cultural and creative environment of European cities to be monitored in the future. Meaning unclear. Suggest 'Although the current update does not delve into the effects of the COVID-19 pandemic and consequent restrictive measures, owing to the lack of data available, developments during the different pandemic phases are likely to be of particular interest in future evolutions of the project.

This report has described the updated methodology in detail and has provided a statistical assessment of the updated results. It also contains a guide for users on the accompanying online platform.

¹⁸ Urban Audit 2021 boundaries have been used. The boundaries are defined based on Dijkstra and Poelman (2012) and are made available by Eurostat at this address: <https://ec.europa.eu/eurostat/web/gisco/geodata/reference-data/administrative-units-statistical-units/urban-audit>.

References

Becker, W., Caperna, G., Del Sorbo, M., Norlen, H., Papadimitriou, E. and Saisana, M. (2022), 'COINr: an R package for developing composite indicators', *Journal of Open Source Software*, Vol. 7, No 78, 4567, doi:10.21105/joss.04567.

Caperna, G. (2022), *JRC Statistical Audit of Africa Regional Integration Index 2019*, Publications Office of the European Union, Luxembourg, doi:10.2760/291502, JRC131063.

Dijkstra, L. and Poelman, H. (2012), *Cities in Europe: The new OECD-EC definition*, European Commission, Brussels.

Eurostat (2016), *Urban Europe – Statistics on cities, towns and suburbs: 2016 edition*, Publications Office of the European Union, Luxembourg, doi:10.2785/91120.

Jenks, G. F. (1967), 'The data model concept in statistical mapping', *International Yearbook of Cartography*, Vol. 7, pp. 186–190.

Montalto, V., Tacao Moura, C., Panella, F., Alberti, V., Becker, W. and Saisana, M. (2019), *The Cultural and Creative Cities Monitor – 2019 edition*, Publications Office of the European Union, Luxembourg, doi:10.2760/257371, JRC117336.

OECD and JRC (Organisation for Economic Co-operation and Development and European Commission, Joint Research Centre) (2008), *Handbook on Constructing Composite Indicators– Methodology and user guide*, OECD Publishing, Paris.

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Annexes

Annex 1. Full list of cities included in the 2023 update of the monitor

City	Country	City	Country	City	Country	City		Country
Graz	AT	Veszprém	HU	Prešov	SK	Hamburg		DE
Linz	AT	Cork	IE	Trenčín	SK	Karlsruhe		DE
Antwerp	BE	Dublin	IE	Ghent	BE	Leipzig		DE
Bruges	BE	Galway	IE	Kortrijk	BE	Mainz		DE
Brussels	BE	Limerick	IE	Brno	CZ	Munich		DE
Mons	BE	Waterford	IE	Hannover	DE	Nuremberg		DE
Plovdiv	BG	Bologna	IT	Heidelberg	DE	Stuttgart		DE
Sofia	BG	Cagliari	IT	Mannheim	DE	Odense		DK
Varna	BG	Florence	IT	Tartu	EE	Lleida		ES
Veliko Tarnovo	BG	Genoa	IT	Barcelona	ES	San Cristóbal de La Laguna (*)		ES
Nicosia	CY	Lecce	IT	Bilbao	ES	Santa Cruz de Tenerife		ES
Ostrava	CZ	Matera	IT	Granada	ES	Valencia		ES
Pilsen	CZ	Perugia	IT	Seville	ES	Espoo		FI
Prague	CZ	Ravenna	IT	Terrassa (*)	ES	Kuopio		FI
České Budějovice	CZ	Kaunas	LT	Valladolid	ES	Tampere		FI
Berlin	DE	Klaipeda	LT	Limoges	FR	Montpellier		FR
Chemnitz	DE	Vilnius	LT	Rouen	FR	Nantes		FR
Essen	DE	Luxembourg	LU	Saint-Etienne	FR	Split		HR
Weimar	DE	Liepāja	LV	Budapest	HU	Zagreb		HR
Aarhus	DK	Riga	LV	Bergamo	IT	Kecskemét		HU
Copenhagen	DK	Valletta	MT	Milan	IT	Miskolc		HU
Tallinn	EE	Amsterdam	NL	Modena	IT	Szeged		HU
Athens	EL	Eindhoven	NL	Parma	IT	Brescia		IT
Kalamata	EL	Leeuwarden	NL	Pesaro	IT	Catania		IT
Patras	EL	Maastricht	NL	Rome	IT	Como		IT
Thessaloniki	EL	Rotterdam	NL	Turin	IT	Naples		IT
Burgos	ES	Bergen	NO	Utrecht	NL	Pordenone		IT
Cordova	ES	Stavanger	NO	Gdynia	PL	Trento		IT
Las Palmas	ES	Gdańsk	PL	Łódź	PL	Trieste		IT
Madrid	ES	Katowice	PL	Braga	PT	Venice		IT
Salamanca	ES	Kraków	PL	Norrköping	SE	's-Hertogenbosch		NL
San Sebastián-Donostia	ES	Lublin	PL	Ljubljana	SI	Amersfoort		NL
Santiago	ES	Warsaw	PL	Salzburg	AT	Groningen		NL
Zaragoza	ES	Wrocław	PL	Vienna	AT	Leiden		NL
Helsinki	FI	Aveiro	PT	Leuven	BE	The Hague		NL
Oulu	FI	Guimarães	PT	Liège	BE	Tilburg		NL
Turku	FI	Lisbon	PT	Namur	BE	Oslo		NO
Avignon	FR	Ponta Delgada	PT	Ostend	BE	Poznań		PL
Bordeaux	FR	Porto	PT	Stara Zagora	BG	Toruń		PL
Lille	FR	Baia Mare	RO	Basel	CH	Coimbra		PT
Lyon	FR	Bucharest	RO	Bern	CH	Faro		PT
Marseille	FR	Cluj-Napoca	RO	Geneva	CH	Sintra		PT
Paris	FR	Sibiu	RO	Zurich	CH	Viseu		PT
Toulouse	FR	Timișoara	RO	Karlovy Vary	CZ	Iași		RO
Osijek	HR	Lund	SE	Olomouc	CZ	Gothenburg		SE
Pula	HR	Stockholm	SE	Bochum	DE	Malmö		SE
Rijeka	HR	Umeå	SE	Bremen	DE	Uppsala		SE
Debrecen	HU	Maribor	SI	Cologne	DE	Bratislava		SK
Győr	HU	Košice	SK	Dresden	DE	1. European Capital of Culture	2. UNESCO Creative City	3. Festivals
Pécs	HU	Nitra	SK	Frankfurt	DE			

(*) San Cristóbal de La Laguna is within the boundaries and analysis for Santa Cruz de Tenerife, and Terrassa is within the boundaries and analysis for Barcelona.

Annex 2. Cities not included in the 2023 update of the monitor

The following cities are not included in the 2023 update of the monitor, despite fulfilling the inclusion criteria.

Inclusion criterion	City	Reason for exclusion
Festivals	Limassol	Not in Urban Audit any more
UNESCO Creative City	Terrassa	Part of Barcelona
Festivals	San Cristóbal de La Laguna	Part of Santa Cruz de Tenerife
Festivals	London	United Kingdom
Festivals	Birmingham	United Kingdom
European Capital of Culture	Glasgow	United Kingdom
UNESCO Creative City	Bradford	United Kingdom
European Capital of Culture	Liverpool	United Kingdom
UNESCO Creative City	Edinburgh	United Kingdom
UNESCO Creative City	Manchester	United Kingdom
UNESCO Creative City	Nottingham	United Kingdom
UNESCO Creative City	York	United Kingdom
UNESCO Creative City	Dundee	United Kingdom
UNESCO Creative City	Norwich	United Kingdom
Festivals	Leeds	United Kingdom
UNESCO Creative City	Bristol	United Kingdom
Festivals	Brighton and Hove	United Kingdom
UNESCO Creative City	Belfast	United Kingdom
Festivals	Portsmouth	United Kingdom
Festivals	Worcester	United Kingdom
Festivals	Bath and North East Somerset	United Kingdom
Festivals	Cheltenham	United Kingdom

Annex 3. Indicators framework

Dimension	Indicator	Short description	Geographical level	Source
Cultural Venues & Facilities	Sights & landmarks	Number of points of historical, cultural and or artistic interest, such as architectural buildings, religious sites, monuments and statues, churches and cathedrals, bridges, towers and fountains, divided by the total population and then multiplied by 100 000.	City	TripAdvisor
	Museums & art galleries	Number of museums and art galleries that are open to the public divided by the total population and then multiplied by 100 000.	City	TripAdvisor
	Cinemas	Number of cinemas in the city divided by the total population and then multiplied by 100 000.	City	OpenStreetMap
	Concert & music halls	Number of theatres and other music venues (concert halls, clubs, etc.) divided by the total population and then multiplied by 100 000.	City	TripAdvisor
	Theatres	Number of theatres in the city divided by the total population and then multiplied by 100 000.	City	OpenStreetMap
Cultural Participation & Attractiveness	Tourist overnight stays	Total annual number of nights that tourists/guests have spent in tourist accommodation establishments (hotel or similar) divided by the total population.	City	Eurostat (Urban Audit)
	Museum visitors	Total number of museum tickets sold during the reference year divided by the total population and then multiplied by 1 000.	City	Eurostat (Urban Audit)
	Cinema attendance	Total number of tickets sold, referring to all films screened during the year, divided by the total population and then multiplied by 1 000.	City	Eurostat (Urban Audit)
	Satisfaction with cultural facilities	Percentage of population that is very satisfied with cultural facilities in the city.	City	Flash Eurobarometer 366 survey: 'Quality of Life in European Cities'
Creative & Knowledge-based Jobs	Jobs in arts, culture & entertainment	Number of jobs in arts-, culture- and entertainment-related activities, such as performing arts, museums and libraries, divided by the total population and then multiplied by 1 000 (NACE Rev. 2, R-U).	City	Eurostat (Urban Audit)

	Jobs in media & communication	Number of jobs in media- and communication-related activities, such as book and music publishing, film and television production , divided by the total population and then multiplied by 1 000 (NACE Rev. 2, J).	City	Eurostat (Urban Audit)
	Jobs in other creative sectors	Number of jobs in professional, scientific and technical, administrative and support service activities, such as architecture, advertising, design and photographic activities, divided by the total population and then multiplied by 1 000 (NACE Rev. 2, M-N).	City	Eurostat (Urban Audit)
Intellectual Property & Innovation	ICT patent applications	Three-year average number of ICT patent applications (including consumer electronics, computers and office machinery, and telecommunications) filed to the European Patent Office by priority year, divided by the total population and then multiplied by 1 million.	NUTS 3	Organisation for Economic Co-operation and Development Regpat database
	Gross expenditure in research & development	Research and development expenditure in business enterprise, government, higher education, private non-profit (aggregated) as a proportion of GDP.	NUTS 2	Eurostat (regional statistics)
New Jobs in Creative Sectors	Jobs in new arts, culture & entertainment enterprises	Number of persons employed in the enterprises established in the reference year in arts, culture and entertainment activities, such as performing arts, museums and libraries, divided by the total population and then multiplied by 100 000.	NUTS 3	Eurostat (regional statistics)
	Jobs in new media & communication enterprises	Number of persons employed in the enterprises established in the reference year in media and communication activities, such as book and music publishing, film and television production, divided by the total population and then multiplied by 100 000.	NUTS 3	Eurostat (regional statistics)
	Jobs in new enterprises in other creative sectors	Number of persons employed in the enterprises established in the reference year in professional, scientific and technical activities, such as architecture, advertising, design and photographic activities, divided by the total population and then multiplied by 100 000.	NUTS 3	Eurostat (regional statistics)
Human Capital & Education	Graduates in arts & humanities	Number of tertiary education graduates (ISCED 2011 levels 5–8) in arts and humanities courses in the reference year, divided by the total population and then multiplied by 100 000.	City	ETER
	Graduates in ICT	Number of tertiary education graduates (ISCED 2011 levels 5–8) in ICT courses in the reference year, divided by the total population and then multiplied by 100 000.	City	ETER

Openness, Tolerance & Trust	Foreign graduates	Number of foreign graduates enrolled in tertiary education courses in the reference year, divided by the total number of tertiary education graduates (ISCED 2011 levels 5–8) in the same academic year.	City	ETER
	Foreign-born population	Percentage of the total population who is foreign-born.	City	Eurostat (Urban Audit)
	Tolerance of foreigners	Percentage of the population who very strongly agrees with the statement 'The presence of foreigners is good for this city'.	City	Flash Eurobarometer 366 survey: 'Quality of Life in European Cities'
	Integration of foreigners	Percentage of the population who very strongly agrees with the statement 'Foreigners who live in this city are well integrated'.	City	Flash Eurobarometer 366 survey: 'Quality of Life in European Cities'
	People trust	Percentage of the population who very strongly agrees with the statement: 'Generally speaking, most people in this city can be trusted'.	City	Flash Eurobarometer 366 survey: 'Quality of Life in European Cities'
Local & International Connections	Passenger flights	Population-weighted average number of accessible passenger flights per day, within 1 hour 30 minutes of travel by road.	City	Directorate-General for Regional and Urban Policy
	Potential road accessibility	Population accessible within 1 hour 30 minutes by road, as share of the population in a neighbourhood of 120 km radius.	City	Directorate-General for Regional and Urban Policy
	Direct trains to other cities	Population accessible within 1 hour 30 minutes by rail (average total travel time), as share of the population in a neighbourhood of 120 km radius.	City	Directorate-General for Regional and Urban Policy
Quality of Governance	Quality of governance	Computed indicator measuring the quality of government in three areas of public services: education, healthcare and law enforcement.	NUTS 2, NUTS 1 and NUTS 0	Directorate-General for Regional and Urban Policy

NB: Indicators shaded in grey have not been updated or included in the 2023 update of the monitor due to low data coverage.

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